

Spring Data JPA with Spring Boot, Hibernate

1) Spring Data JPA - Quick Example

Expected Output

After running OrmLearnApplication, the console displays:

INFO Inside main

INFO Start

DEBUG countries=[Country [code=IN, name=India], Country [code=US, name=United States of America]]

INFO End

The data fetched from the database is:

Code Name

IN India

US United States of America

2) Difference between JPA, Hibernate and Spring Data JPA

Comparison Table

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	Framework / ORM Tool	Framework
Developed By	Java (Jakarta/Oracle)	Hibernate Team	Spring
Implementation	No implementation	Implements JPA	Uses JPA implementations
Purpose	Defines persistence standards	Provides ORM functionality	Reduces boilerplate code
SQL Generation	Not directly	Yes	Through Hibernate/JPA
Transaction Management	Not provided	Manual or JPA based	Automatic with @Transactional
Boilerplate Code	More	Moderate	Very Less

Feature	JPA	Hibernate	Spring Data JPA
Repository Support	No	No	Yes (JpaRepository)

Without Hibernate / JPA

```
Connection con = DriverManager.getConnection(...);
```

```
PreparedStatement ps =
con.prepareStatement(
    "insert into employee values (?, ?)");
```

```
ps.setInt(1, 1);
ps.setString(2, "John");
```

```
ps.executeUpdate();
```

Expected Output

Employee record inserted successfully.

Using Hibernate

```
Session session = factory.openSession();
```

```
Transaction tx = session.beginTransaction();
```

```
session.save(employee);
```

```
tx.commit();
```

```
session.close();
```

Expected Output

Employee saved successfully using Hibernate.

Using Spring Data JPA

Repository

```
public interface EmployeeRepository  
    extends JpaRepository<Employee, Integer> {  
}
```

Service

```
@Autowired  
private EmployeeRepository employeeRepository;  
  
@Transactional  
public void addEmployee(Employee employee) {  
    employeeRepository.save(employee);  
}
```

Expected Output

Employee saved successfully using Spring Data JPA.